Inserting Missing Numbers — Achievers / HOTS — Answer Key

1. **B) 3** — Each column adds up to 20 (col 1: 9+6+5=20, col 2: 4+10+6=20). For column 3: 7 + ? + 10 = 20, so ? = 20 − 17 = 3.
2. **A) 10** — Each column adds up to 24 (col 1: 8+9+7=24, col 3: 6+11+7=24). For column 2: ? + 5 + 9 = 24, so ? = 24 − 14 = 10.
3. **C) 18** — Each row follows (col1 + col2) × 2 = col3: (3+4)×2=14, (5+2)×2=14. For row 3: (6+3)×2 = 18.
4. **D) 3** — Each row follows (2 × col1) + col2 = col3: (2×3)+4=10, (2×5)+2=12. For row 3: (2×6) + ? = 15, so 12 + ? = 15, ? = 3.
5. **A) 13** — Each row follows (col1 − col2) × 3 = col3: (9−4)×3=15, (11−3)×3=24. For row 3: (? − 6) × 3 = 21, so ? − 6 = 7, ? = 13.
6. **D) 5** — Every row and column adds up to 15. Using row 2: 9 + ? + 1 = 15, so ? = 15 − 10 = 5. (Check column 2: 7 + 5 + 3 = 15 — it works both ways!)
7. **C) 30** — The two middle numbers are 7+9=16 and 9+5=14. The top number is 16+14 = 30.
8. **C) 5** — Top = Left + 2×Middle + Right: 3+8+5=16, 6+4+7=17. For row 3: 5 + 2×? + 6 = 21, so 2×? = 10, ? = 5.
9. **D) 7** — Top = Left + 2×Middle + Right: 2+6+4=12, 5+4+6=15. For row 3: ? + 10 + 3 = 20, so ? = 20 − 13 = 7.
10. **D) 12** — Top-left × bottom-right = top-right × bottom-left: 4 × ? = 6 × 8 = 48, so ? = 48 ÷ 4 = 12.
11. **A) 5** — Each column adds up to 27 (col 1: 10+8+9=27, col 2: 12+7+8=27). For column 3: ? + 12 + 10 = 27, so ? = 27 − 22 = 5.
12. **D) 38** — Each row follows (col1 + col2) × 2 = col3: (8+5)×2=26, (10+3)×2=26. For row 3: (12+7)×2 = 38.
13. **D) 51** — The two middle numbers are 12+15=27 and 15+9=24. The top number is 27+24 = 51.
14. **D) 7** — Every row and column adds up to 22. Using column 1: 9 + 6 + ? = 22, so ? = 22 − 15 = 7. (Check row 3: 7 + 4 + 11 = 22 — it works both ways!)
15. **B) 6** — Each row follows (3 × col1) − col2 = col3: (3×4)−5=7, (3×6)−8=10. For row 3: (3×5) − ? = 9, so 15 − ? = 9, ? = 6.

How to use this: 10+/15 is strong at this difficulty. For the magic-square questions, check your answer against BOTH its row and its column — a true magic-square answer always agrees with both.